

Differential Equations Notes: Laplace Transforms & Inverse Laplace Transforms

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1 Introduction to Laplace Transform (IMG_1023)

Transforming from domain $t \rightarrow s$:

$$\begin{aligned}\frac{dy}{dx} + 3y = 0 &\xrightarrow{\mathcal{L}} y(s) \xrightarrow{\mathcal{L}^{-1}} y(x) \\ s + 3 = 0 &\implies s = -3\end{aligned}$$

Definition of Laplace Transform

The Laplace Transform of a function $f(t)$ defined for $t \geq 0$ is:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

Conditions for Existence

The Laplace transform $\mathcal{L}\{f(t)\}$ exists if:

1. $f(t)$ is **piecewise continuous** on every finite interval in $[0, \infty)$.
2. $f(t)$ is of **exponential order** k as $t \rightarrow \infty$, i.e., $|f(t)| \leq Me^{kt}$ for all $t \geq T$.

Examples of Order:

- $\sin t, \cos t, e^{kt} \implies$ Exponential order
- $e^{t^2} \implies$ Not of exponential order

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Standard Transform 1: $\mathcal{L}\{1\}$

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} \cdot 1 dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = 0 - \left(-\frac{1}{s} \right) = \frac{1}{s} \quad (s > 0)$$

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2 Standard Formulas of Laplace Transforms (IMG_1024 & IMG_1025)

2. Transform of Power Functions: $\mathcal{L}\{t^n\}$

Using the substitution $u = st \implies dt = \frac{du}{s}$:

$$\begin{aligned}\mathcal{L}\{t^n\} &= \int_0^\infty e^{-st} t^n dt = \int_0^\infty e^{-u} \left(\frac{u}{s}\right)^n \frac{du}{s} \\ &= \frac{1}{s^{n+1}} \int_0^\infty e^{-u} u^n du = \frac{\Gamma(\mathbf{n} + 1)}{\mathbf{s}^{n+1}}\end{aligned}$$

3. For non-negative integers n :

$$\mathcal{L}\{t^n\} = \frac{\mathbf{n}!}{\mathbf{s}^{n+1}}$$

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4. Transform of Exponential Function: $\mathcal{L}\{e^{at}\}$

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st} e^{at} dt = \int_0^\infty e^{-(s-a)t} dt = \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^\infty = \frac{1}{\mathbf{s} - \mathbf{a}} \quad (s > a)$$

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5. Transform of Cosine: $\mathcal{L}\{\cos at\}$

$$\mathcal{L}\{\cos at\} = \int_0^\infty e^{-st} \cos at dt = \left[\frac{e^{-st}}{s^2 + a^2} (-s \cos at + a \sin at) \right]_0^\infty = \frac{\mathbf{s}}{\mathbf{s}^2 + \mathbf{a}^2}$$

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6. Transform of Sine: $\mathcal{L}\{\sin at\}$

$$\mathcal{L}\{\sin at\} = \int_0^\infty e^{-st} \sin at dt = \left[\frac{e^{-st}}{s^2 + a^2} (-s \sin at - a \cos at) \right]_0^\infty = \frac{\mathbf{a}}{\mathbf{s}^2 + \mathbf{a}^2}$$

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7. Transform of Hyperbolic Cosine: $\mathcal{L}\{\cosh at\}$

$$\mathcal{L}\{\cosh at\} = \frac{\mathbf{s}}{\mathbf{s}^2 - \mathbf{a}^2} \quad (s > |a|)$$

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8. Transform of Hyperbolic Sine: $\mathcal{L}\{\sinh at\}$

$$\mathcal{L}\{\sinh at\} = \frac{\mathbf{a}}{\mathbf{s}^2 - \mathbf{a}^2} \quad (s > |a|)$$

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3 Properties of Laplace Transform (IMG_1026 & IMG_1027)

First Shifting Property (First Frequency-Shift Theorem)

Let $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

or equivalently for the inverse:

$$\mathcal{L}^{-1}\{F(s - a)\} = e^{at}f(t)$$

Proof:

$$\mathcal{L}\{e^{at}f(t)\} = \int_0^\infty e^{-st}e^{at}f(t) dt = \int_0^\infty e^{-(s-a)t}f(t) dt$$

Let $u = s - a$, then:

$$= \int_0^\infty e^{-ut}f(t) dt = F(u) = F(s - a)$$

Example:

$$\mathcal{L}\{e^{at} \sin bt\} = \frac{b}{(s - a)^2 + b^2}$$

Change of Scale Property

Let $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Proof:

$$\mathcal{L}\{f(at)\} = \int_0^\infty e^{-st}f(at) dt$$

Substitute $u = at \implies t = \frac{u}{a}, dt = \frac{du}{a}$:

$$= \int_0^\infty e^{-s(u/a)}f(u)\frac{du}{a} = \frac{1}{a} \int_0^\infty e^{-\left(\frac{s}{a}\right)u}f(u) du = \frac{1}{a}F\left(\frac{s}{a}\right)$$

Verification using $\sin at$: Since $\mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}$:

$$\mathcal{L}\{\sin at\} = \frac{1}{a} \left[\frac{1}{\left(\frac{s}{a}\right)^2 + 1} \right] = \frac{1}{a} \left[\frac{a^2}{s^2 + a^2} \right] = \frac{a}{s^2 + a^2}$$

4 Applications & Solved Examples (IMG_1028)

Example 1: Evaluating Integrals using Laplace Transform

Evaluate $\int_0^\infty e^{-2t} \sin 4t dt$.

Solution: Recall $\mathcal{L}\{\sin 4t\} = \int_0^\infty e^{-st} \sin 4t \, dt = \frac{4}{s^2+4^2} = \frac{4}{s^2+16}$.

Here $s = 2$, so substituting $s = 2$:

$$\int_0^\infty e^{-2t} \sin 4t \, dt = \frac{4}{2^2 + 16} = \frac{4}{4 + 16} = \frac{1}{5}$$

Example 2: Transform of $2e^{-t} \cos^2\left(\frac{t}{2}\right)$

Find $\mathcal{L}\left\{2e^{-t} \cos^2\left(\frac{t}{2}\right)\right\}$.

Solution: Using the trigonometric identity $2 \cos^2\left(\frac{t}{2}\right) = \cos t + 1$:

$$2e^{-t} \cos^2\left(\frac{t}{2}\right) = e^{-t}(\cos t + 1) = e^{-t} + e^{-t} \cos t$$

Taking Laplace transform:

$$\mathcal{L}\left\{2e^{-t} \cos^2\left(\frac{t}{2}\right)\right\} = \mathcal{L}\{e^{-t}\} + \mathcal{L}\{e^{-t} \cos t\}$$

Using first shifting property ($a = -1$):

$$= \frac{1}{s+1} + \frac{s+1}{(s+1)^2 + 1}$$

5 Inverse Laplace Transform via Partial Fractions (IMG_1029 – IMG_1032)

General Decomposition Rules (IMG_1029)

1. ****Distinct Linear Factors:****

$$Y(s) = \frac{F(s)}{(s-a_1)(s-a_2)\dots(s-a_n)} = \frac{A_1}{s-a_1} + \frac{A_2}{s-a_2} + \dots + \frac{A_n}{s-a_n}$$

2. ****Repeated Linear Factors:****

$$Y(s) = \frac{F(s)}{(s-a)^n} = \frac{A_1}{s-a} + \frac{A_2}{(s-a)^2} + \dots + \frac{A_n}{(s-a)^n}$$

3. ****Distinct Quadratic Factors:****

$$Y(s) = \frac{F(s)}{(a_1s^2 + b_1s + c_1)(a_2s^2 + b_2s + c_2)\dots} = \frac{A_1s + B_1}{a_1s^2 + b_1s + c_1} + \frac{A_2s + B_2}{a_2s^2 + b_2s + c_2} + \dots$$

4. ****Repeated Quadratic Factors:****

$$Y(s) = \frac{F(s)}{(as^2 + bs + c)^n} = \frac{A_1s + B_1}{as^2 + bs + c} + \frac{A_2s + B_2}{(as^2 + bs + c)^2} + \dots + \frac{A_ns + B_n}{(as^2 + bs + c)^n}$$

Example 1: Distinct Linear Factors (IMG_1030)

Find $\mathcal{L}^{-1} \left\{ \frac{6}{(s-4)(s+2)} \right\}$.

Solution: Set up partial fractions:

$$\frac{6}{(s-4)(s+2)} = \frac{A}{s-4} + \frac{B}{s+2}$$

$$6 = A(s+2) + B(s-4) = (A+B)s + (2A-4B)$$

Equating coefficients:

$$A + B = 0 \implies B = -A$$

$$2A - 4B = 6 \implies 2A - 4(-A) = 6 \implies 6A = 6 \implies A = 1, B = -1$$

Now take the inverse transform:

$$\mathcal{L}^{-1} \left\{ \frac{6}{(s-4)(s+2)} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s-4} - \frac{1}{s+2} \right\} = e^{4t} - e^{-2t}$$

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Example 2: Repeated Linear Factors (IMG_1031)

Find $\mathcal{L}^{-1} \left\{ \frac{s}{(s+1)^2} \right\}$.

Solution: Set up partial fractions:

$$\frac{s}{(s+1)^2} = \frac{A}{s+1} + \frac{B}{(s+1)^2}$$

$$s = A(s+1) + B = As + (A+B)$$

Equating coefficients:

$$A = 1, \quad A + B = 0 \implies B = -1$$

Now take the inverse transform:

$$\mathcal{L}^{-1} \left\{ \frac{s}{(s+1)^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s+1} - \frac{1}{(s+1)^2} \right\} = e^{-t} - te^{-t}$$

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Example 3: Quadratic Factor Case (IMG_1031 & IMG_1032)

Find $\mathcal{L}^{-1} \left\{ \frac{s}{s^4+4a^4} \right\}$.

Solution: First factor the denominator $s^4 + 4a^4$:

$$s^4 + 4a^4 = (s^2 + 2a^2)^2 - 4a^2s^2 = (s^2 - 2as + 2a^2)(s^2 + 2as + 2a^2)$$

Set up partial fractions:

$$\frac{s}{s^4 + 4a^4} = \frac{1}{4a} \left[\frac{1}{s^2 - 2as + 2a^2} - \frac{1}{s^2 + 2as + 2a^2} \right]$$

Completing the square in the denominators:

$$s^2 - 2as + 2a^2 = (s - a)^2 + a^2$$

$$s^2 + 2as + 2a^2 = (s + a)^2 + a^2$$

Substituting into the inverse Laplace transform:

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^4 + 4a^4} \right\} = \frac{1}{4a^2} \left[\mathcal{L}^{-1} \left\{ \frac{a}{(s - a)^2 + a^2} \right\} - \mathcal{L}^{-1} \left\{ \frac{a}{(s + a)^2 + a^2} \right\} \right]$$

Using the first shifting theorem for sine:

$$= \frac{1}{4a^2} [e^{at} \sin at - e^{-at} \sin at] = \frac{1}{2a^2} \sinh(at) \sin(at)$$

6 Laplace Transform of Derivatives (IMG_1033 & IMG_1034)

Theorem: First Derivative

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$$

Proof: By definition of the Laplace transform:

$$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t) dt$$

Using integration by parts ($\int u dv = uv - \int v du$ where $u = e^{-st}$ and $dv = f'(t) dt$):

$$= [e^{-st} f(t)]_0^\infty - \int_0^\infty \left(\frac{d}{dt} e^{-st} \right) f(t) dt$$

Assuming $f(t)$ is of exponential order, $\lim_{t \rightarrow \infty} e^{-st} f(t) = 0$:

$$\begin{aligned} &= -f(0) + s \int_0^\infty e^{-st} f(t) dt \\ &= s\mathcal{L}\{f(t)\} - f(0) \end{aligned}$$

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General Formula: n -th Derivative

In general, for the n -th derivative:

$$\mathcal{L}\{f^{(n)}(t)\} = s^n \mathcal{L}\{f(t)\} - s^{n-1}f(0) - s^{n-2}f'(0) - s^{n-3}f''(0) - \dots - f^{(n-1)}(0)$$

Proof by Mathematical Induction (IMG_1033 & IMG_1034): Base Case ($n = 1$):

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} e^{-st} f'(t) dt = s\mathcal{L}\{f(t)\} - f(0) \quad (\text{True})$$

Inductive Hypothesis: Assume the result holds for some integer k :

$$\mathcal{L}\{f^{(k)}(t)\} = s^k \mathcal{L}\{f(t)\} - s^{k-1} f(0) - s^{k-2} f'(0) - \dots - f^{(k-1)}(0)$$

Inductive Step ($n = k + 1$): Let $g(t) = f^{(k)}(t) \implies g'(t) = f^{(k+1)}(t)$. Using the base case formula for $g'(t)$:

$$\begin{aligned} \mathcal{L}\{f^{(k+1)}(t)\} &= \mathcal{L}\{g'(t)\} = s\mathcal{L}\{g(t)\} - g(0) \\ &= s\mathcal{L}\{f^{(k)}(t)\} - f^{(k)}(0) \end{aligned}$$

Substituting the inductive hypothesis:

$$\begin{aligned} &= s [s^k \mathcal{L}\{f(t)\} - s^{k-1} f(0) - s^{k-2} f'(0) - \dots - f^{(k-1)}(0)] - f^{(k)}(0) \\ &= s^{k+1} \mathcal{L}\{f(t)\} - s^k f(0) - s^{k-1} f'(0) - \dots - f^{(k)}(0) \end{aligned}$$

Thus, the formula holds for all integers $n \geq 1$ by mathematical induction. ■

Special Cases (IMG_1034)

1. $\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$
2. $\mathcal{L}\{f''(t)\} = s^2 \mathcal{L}\{f(t)\} - sf(0) - f'(0)$
3. $\mathcal{L}\{f'''(t)\} = s^3 \mathcal{L}\{f(t)\} - s^2 f(0) - sf'(0) - f''(0)$

Example 1: Find $\mathcal{L}\{\sin^2 t\}$ using Derivative Rule (IMG_1034)

Let $f(t) = \sin^2 t \implies f(0) = 0$. Taking derivative:

$$f'(t) = 2 \sin t \cos t = \sin 2t$$

Apply Laplace transform to both sides:

$$\mathcal{L}\{\sin 2t\} = \mathcal{L}\{f'(t)\} = s\mathcal{L}\{\sin^2 t\} - f(0)$$

Since $\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 4}$:

$$\begin{aligned} \frac{2}{s^2 + 4} &= s\mathcal{L}\{\sin^2 t\} - 0 \\ \mathcal{L}\{\sin^2 t\} &= \frac{2}{s(s^2 + 4)} \end{aligned}$$

7 Application to Linear Differential Equations (IMG_1035 – IMG_1037)

Given a second-order linear differential equation with constant coefficients:

$$y'' + ay' + by = r(t), \quad y(0) = y_0, \quad y'(0) = y'_0$$

General Procedure: 1. Apply Laplace Transform to both sides:

$$\mathcal{L}\{y''\} + a\mathcal{L}\{y'\} + b\mathcal{L}\{y\} = \mathcal{L}\{r(t)\}$$

$$[s^2Y(s) - sy(0) - y'(0)] + a[sY(s) - y(0)] + bY(s) = R(s)$$

2. Solve algebraically for $Y(s)$:

$$Y(s)(s^2 + as + b) = R(s) + sy(0) + y'(0) + ay(0)$$

$$Y(s) = \frac{R(s) + sy(0) + y'(0) + ay(0)}{s^2 + as + b}$$

3. Apply Inverse Laplace Transform:

$$y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

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Example 1 (IMG_1035 & IMG_1036)

Solve the initial value problem:

$$y' + 3y = 10 \sin t, \quad y(0) = 0$$

Step 1: Apply Laplace Transform

$$\mathcal{L}\{y' + 3y\} = \mathcal{L}\{10 \sin t\}$$

$$[sY(s) - y(0)] + 3Y(s) = 10 \cdot \frac{1}{s^2 + 1}$$

Since $y(0) = 0$:

$$Y(s)(s + 3) = \frac{10}{s^2 + 1} \implies Y(s) = \frac{10}{(s + 3)(s^2 + 1)}$$

Step 2: Partial Fractions

$$\frac{10}{(s + 3)(s^2 + 1)} = \frac{A}{s + 3} + \frac{Bs + C}{s^2 + 1}$$

$$10 = A(s^2 + 1) + (Bs + C)(s + 3) = (A + B)s^2 + (3B + C)s + (A + 3C)$$

Equating coefficients:

$$A + B = 0 \implies A = -B$$

$$3B + C = 0 \implies C = -3B$$

$$A + 3C = 10 \implies (-B) + 3(-3B) = 10 \implies -10B = 10 \implies B = -1$$

So $A = 1$ and $C = 3$.

$$Y(s) = \frac{1}{s + 3} + \frac{-s + 3}{s^2 + 1}$$

Step 3: Inverse Laplace Transform (IMG_1037)

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s+3} \right\} - \mathcal{L}^{-1} \left\{ \frac{s}{s^2+1} \right\} + 3\mathcal{L}^{-1} \left\{ \frac{1}{s^2+1} \right\}$$
$$\mathbf{y}(\mathbf{t}) = \mathbf{e}^{-3\mathbf{t}} - \cos \mathbf{t} + 3 \sin \mathbf{t}$$

Example 2 (IMG_1037)

Solve the initial value problem:

$$y'' + 2y' + y = e^{-t}, \quad y(0) = -1, \quad y'(0) = 1$$

Step 1: Apply Laplace Transform

$$[s^2Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + Y(s) = \frac{1}{s+1}$$

Substitute $y(0) = -1$ and $y'(0) = 1$:

$$[s^2Y(s) + s - 1] + 2[sY(s) + 1] + Y(s) = \frac{1}{s+1}$$

$$(s^2 + 2s + 1)Y(s) + s + 1 = \frac{1}{s+1}$$

$$(s+1)^2Y(s) = -(s+1) + \frac{1}{s+1}$$

$$Y(s) = -\frac{1}{s+1} + \frac{1}{(s+1)^3}$$

Step 2: Inverse Laplace Transform Using shifting theorem $\mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^3} \right\} = e^{-t} \frac{t^2}{2!}$:

$$\mathbf{y}(\mathbf{t}) = -\mathbf{e}^{-\mathbf{t}} + \mathbf{e}^{-\mathbf{t}} \frac{\mathbf{t}^2}{2}$$

8 Laplace Transform of Integrals (IMG_1038)

Theorem

If $\mathcal{L}\{f(t)\} = F(s)$, then:

$$\mathcal{L} \left\{ \int_0^t f(\tau) d\tau \right\} = \frac{F(s)}{s}$$

Equivalently:

$$\mathcal{L}^{-1} \left\{ \frac{F(s)}{s} \right\} = \int_0^t f(\tau) d\tau$$

Example: Find $f(t)$ given $\mathcal{L}\{f(t)\} = \frac{1}{s(s^2+\omega^2)}$

Solution: We know:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+\omega^2}\right\} = \frac{1}{\omega} \sin \omega t$$

Using the Integral Theorem:

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s}\left(\frac{1}{s^2+\omega^2}\right)\right\} &= \int_0^t \frac{1}{\omega} \sin(\omega\tau) d\tau \\ &= \frac{1}{\omega} \left[-\frac{\cos \omega\tau}{\omega}\right]_0^t = \frac{1}{\omega^2} [-\cos \omega t - (-1)] \\ &= \frac{1}{\omega^2} (1 - \cos \omega t)\end{aligned}$$